

Layer-by-Layer Click Deposition of Functional Polymer Coatings for Combating Marine Biofouling

Wen Jing Yang

NUS Graduate School for Integrative Science and Engineering, National University of Singapore, Kent Ridge, Singapore 117576

Dicky Pranantyo, Koon-Gee Neoh, and En-Tang Kang*

Department of Chemical & Biomolecular Engineering, National University of Singapore, Kent Ridge, Singapore, 119260

Serena Lay-Ming Teo*

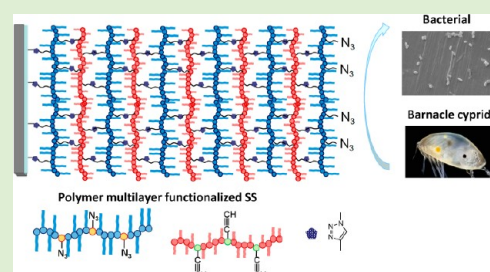
Tropical Marine Science Institute, National University of Singapore, Kent Ridge, Singapore, 119223

Daniel Rittschof*

Nicholas School of the Environment, Duke University Marine Laboratory, 135 Duke Marine Lab Road Beaufort, North Carolina 28516-9721, United States

Supporting Information

ABSTRACT: “Click” chemistry-enabled layer-by-layer (LBL) deposition of multilayer functional polymer coatings provides an alternative approach to combating biofouling. Fouling-resistant azido-functionalized poly(ethylene glycol) methyl ether methacrylate-based polymer chains (azido-poly(PEGMA)) and antimicrobial alkynyl-functionalized 2-(methacryloyloxy)ethyl trimethyl ammonium chloride-based polymer chains (alkynyl-poly(META)) were click-assembled layer-by-layer via alkyne–azide 1,3-dipolar cycloaddition. The polymer multilayer coatings are resistant to bacterial adhesion and are bactericidal to marine Gram-negative *Pseudomonas* sp. NCIMB 2021 bacteria. Settlement of barnacle (*Amphibalanus* (=Balanus) amphitrite) cyprids is greatly reduced on the multilayer polymer-functionalized substrates. As the number of the polymer layers increases, efficacy against bacterial fouling and settlement of barnacle cyprids increases. The LBL-functionalized surfaces exhibit low toxicity toward the barnacle cyprids and are stable upon prolonged exposure to seawater. LBL click deposition is thus an effective and potentially environmentally benign way to prepare antifouling coatings.



1. INTRODUCTION

Biofouling, caused by accumulation of organisms on man-made surfaces, is a worldwide problem affecting marine and aquatic industries.^{1,2} Biocidal coatings, which release copper and organic biocides, have been traditionally used to control aquatic biofouling.^{3,4} However, these coatings are under scrutiny today due to environmental concern about the released toxins and their detrimental effect on aquatic organisms.^{3,4} Consequently, there is an increasing need to develop environmentally benign technologies for inhibiting biofouling.

Recently, tethering of nontoxic functional polymer brush coatings has been suggested as an environmentally friendly approach to prevent the attachment of proteins, bacteria, and marine organisms to surfaces.^{5–7} The sequential buildup of polymers via the layer-by-layer (LBL) technique provides an efficient and versatile means for depositing functional polymer

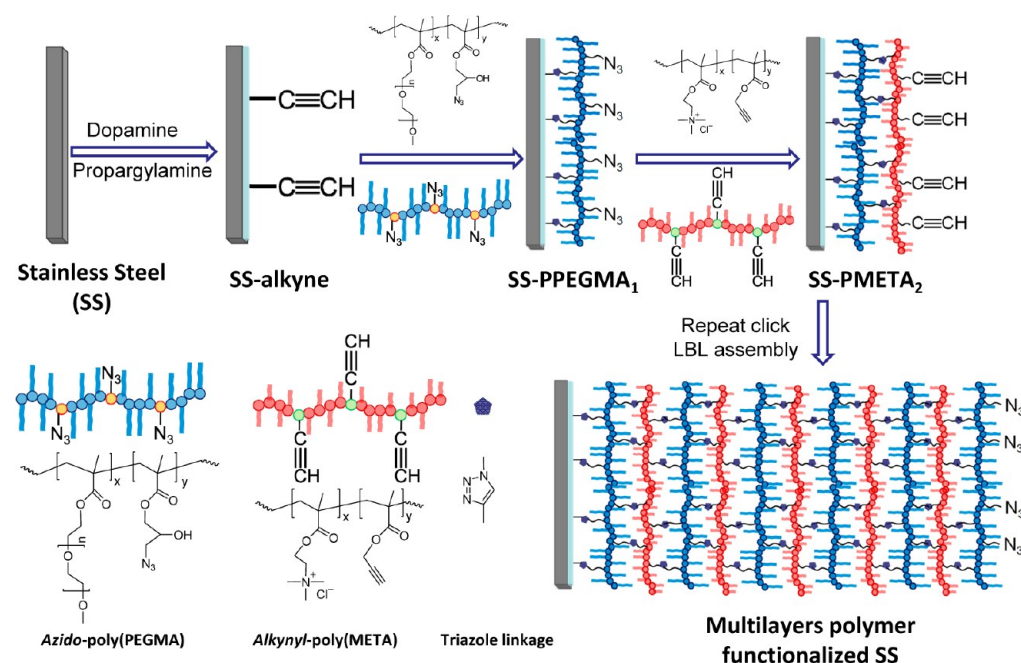
coatings on surfaces.^{8–16} The multilayers are usually assembled through noncovalent electrostatic and hydrogen-bonding interactions. Covalently cross-linked multilayers offer better stability under varying environmental conditions, such as changing pH and salt concentrations, than those formed via electrostatic or hydrogen-bonding interactions.^{9,10} “Click” chemistry, employing the highly efficient Huisgen alkyne–azide cycloaddition reaction, provides a facile means for covalent coupling.^{17–19} Consequently, alkyne–azide click chemistry-enabled LBL assembly provides an effective means for the fabrication of polymer multilayers under mild reaction conditions.^{20–24} The resultant cyclic 1,2,3-triazole linkages between the adjacent layers are resistant to hydrolysis,

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Scheme 1. Schematic Illustration of the Preparation of Antifouling and Antibacterial Polymer Multilayer Coatings via LBL Click Deposition



oxidation, and reduction, making the click reaction of particular interest in the preparation of highly robust multilayers.

To develop an effective coating, it is desirable to combine the antiadhesive and antimicrobial properties simultaneously.^{25,26} Poly(ethylene glycol) (PEG) and its derivatives exhibit good antifouling effects to a wide variety of proteins, and reduce cell attachment and growth.^{27–30} However, they cannot be assembled by the LBL technique through electrostatic interactions due to their uncharged nature. However, low-fouling PEG multilayers can be prepared by LBL click assembly of PEG with clickable functionalities.³⁰ Quaternary ammonium cations (QAC) exhibit antimicrobial activity against bacteria by disrupting their cellular membrane.^{31–34}

In this work, an alternative approach, employing the covalent LBL click deposition of PEG and QAC-containing polymers, is developed to achieve the nonadhesive and antimicrobial polymer coatings (Scheme 1). Propargylamine was coupled to a polydopamine-coated substrate surface via the Schiff-base and Michael addition reaction to provide the alkyne functionality for the subsequent LBL click assembly. Then, *azido*-functionalized poly(ethylene glycol) methyl ether methacrylate-based polymer chains (*azido*-poly(PEGMA)) and *alkynyl*-functionalized 2-(methacryloyloxy)ethyl trimethyl ammonium chloride-based polymer chains (*alkynyl*-poly(META)) were click deposited sequentially on the surface. The antifouling efficiency of the resulting polymer multilayer coatings was assayed by bacterial (*Pseudomonas* sp. NCIMB 2021) adhesion and settlement of barnacle (*Amphibalanus* (=Balanus) amphitrite) cyprids.

2. MATERIALS AND METHODS

2.1. Materials. AISI type 304 stainless steel foils 0.05 mm in thickness were purchased from Goodfellow, Ltd., of Cambridge, UK. Poly(ethylene glycol) methyl ether methacrylate macromonomer (PEGMA, $n \sim 8.5$, average molecular weight $M_n \sim 475$), 2-(methacryloyloxy)ethyl trimethyl ammonium chloride solution (META, 80 wt % in H₂O), glycidyl methacrylate (GMA, 97%), 4-

cyano-4-(phenylcarbonothioylthio)pentanoic acid (97%), sodium azide (NaN₃, 99.5%), (+)-sodium L-ascorbate (98%), propargylamine (98%), dopamine hydrochloride, and copper(II) sulfate pentahydrate (98%) were purchased from Sigma-Aldrich Chemical Co., St. Louis, MO, USA. Propargyl methacrylate (PMA, 98%) was obtained from Alfa Aesar Co., Ward Hill, MA, USA. The monomers, PEGMA, PMA, and GMA, were passed through an inhibitor-removal column (Sigma-Aldrich) and then stored under an argon atmosphere at -10°C . A marine Gram-negative bacterial strain of *Pseudomonas* sp. NCIMB 2021 was obtained from the National Collection of Marine Bacteria, Sussex, U.K. The LIVE/DEAD BacLight Bacterial Viability Kit L131152 was purchased from Molecular Probes Inc., Eugene, OR.

2.2. Polymer Synthesis. The poly(poly(ethylene glycol) methyl ether methacrylate-*co*-glycidyl methacrylate) copolymer, poly-(PEGMA-*co*-GMA), was synthesized via the reversible addition–fragmentation chain transfer polymerization (RAFT) of PEGMA and GMA, as follows: PEGMA (4.99 g, 10.5 mmol), GMA (0.64 g, 4.5 mmol), 4-cyano-4-(phenylcarbonothioylthio)pentanoic acid (21.3 mg, 0.075 mmol), and azobisisobutyronitrile (AIBN, 2.5 mg, 0.015 mmol) were introduced into 10 mL of 1,4-dioxane in a reaction flask. The mixture was purged with purified argon for 30 min and then heated to 70°C under stirring. After 12 h, the reaction mixture was exposed to air to stop the polymerization, and the copolymer was precipitated in 200 mL of diethyl ether. The copolymer was purified twice by redissolving in 1,4-dioxane and reprecipitating in diethyl ether. The poly(PEGMA-*co*-GMA) copolymer with azide functionalities (*azido*-poly(PEGMA)) was synthesized by reaction of sodium azide with the pendant oxirane rings in the GMA units of the random copolymer.³⁵ Sodium azide (NaN₃, 0.88 g, 13.5 mmol) and ammonium chloride (NH₄Cl, 0.72 g, 13.5 mmol) were added to a 50 mL aqueous solution of poly(PEGMA-*co*-GMA) (4.8 g, containing about 4.5 mmol oxirane moieties). The reaction mixture was stirred at 50°C for 24 h. After the reaction, the mixture was dialyzed against deionized water for 3 days to remove the excess NaN₃ and NH₄Cl. The resulting *azido*-poly-(PEGMA) was dried by lyophilization. Gel permeation chromatography (GPC): $M_n = 85344\text{ g}\cdot\text{mol}^{-1}$; polydispersity index (PDI): 1.10.

The poly(2-(methacryloyloxy)ethyl trimethyl ammonium chloride-*co*-propargyl methacrylate) copolymer with alkyne functionalities [poly(META-*co*-PMA)] was synthesized by conventional free radical copolymerization. META (2.84 mL, 12 mmol), PMA (0.38 mL, 3.0 mmol), and AIBN (5 mg, 0.03 mmol) were introduced into a reaction

flask containing 10 mL of ethanol. The solution was purged with purified argon for 30 min and then placed in an oil bath at 65 °C. After 1 h, the copolymer was recovered by precipitation into isopropanol and purified twice by redissolving in ethanol and reprecipitating in isopropanol. The resulting copolymer is denoted as *alkynyl-poly-(META)*. As the alkynyl hydrogen is quite labile under radical polymerization conditions, the polymerization was carefully controlled to avoid the occurrence of cross-linking.^{36,37} The *alkynyl-poly-(META)* copolymer was prepared by using a very short polymerization time (1 h). GPC: $M_n = 28160 \text{ g}\cdot\text{mol}^{-1}$; PDI: 2.15.

2.3. Polymer Multilayer Coatings Prepared by LBL Click Assembly. Polydopamine-coated stainless steel (SS-PDA) substrates were prepared as reported in the literature.^{38,39} Alkyne-functionalized SS substrates were prepared by the Schiff-base and Michael addition reaction of propargylamine with SS-PDA.³⁸ SS-PDA substrates were immersed in the propargylamine solution (2 mg/mL) at room temperature for 6 h. The resulting propargylamine-coated SS (SS-alkyne) substrates were sequentially exposed to alternate solutions of *azido-poly(PEGMA)* and *alkynyl-poly(META)*, containing copper(II) sulfate and sodium ascorbate, at room temperature for 1 h.^{20–22} The coating solutions consisted of 3:1:1 volume ratio of *azido-poly(PEGMA)* (or *alkynyl-poly(META)*) (2 mg/mL, 9 mL), copper(II) sulfate (0.72 mg/mL, 3 mL), and sodium ascorbate (1.76 mg/mL, 3 mL). Typically, a total of six bilayers can be completed in 1 day. After the alternating layers were click deposited, the substrates were washed with copious amounts of deionized water. To prevent oxidation of the copper, copper(II) sulfate solutions were prepared afresh prior to the deposition of each layer. The multilayer polymer-functionalized surfaces were denoted as the SS-PPEMGA_x or SS-PMETA_x, where x represents the number of layers for each type of the polymer clicked. In addition, 0.1 mL of a mixture of *azido-poly(PEGMA)* (2 mg/mL) and *alkynyl-poly(META)* (2 mg/mL) was spin-coated on the SS-alkyne surface. Then, 0.1 mL of copper(II) sulfate (0.72 mg/mL) and sodium ascorbate (1.76 mg/mL) was dropped on the substrate for 1 h to trigger the click cross-linking reaction. The resulting substrate was denoted as the SS-Mix surface and tested in the bacterial fouling assays. All the polymer-functionalized surfaces were washed thoroughly with deionized water under ultrasonication to eliminate the residual copper catalyst. Glass vials (8 mL) were used as the substrates for barnacle cyprid settlement assays. The process for preparing polymer multilayer coatings on the inside surface of glass vials was the same as that for the stainless steel substrates.

2.4. Polymer Characterization. Fourier transform infrared (FTIR) spectra of the copolymers were recorded on a Bio-Rad FTS 135 FT-IR spectrophotometer, with the copolymer samples dispersed in KBr pellets. Each spectrum was collected by cumulating 32 scans at a resolution of 16 cm^{-1} and the diffuse reflectance spectra were scanned over the range of 500–4000 cm^{-1} . Proton nuclear magnetic resonance (^1H NMR) spectra of the copolymers were measured on a Bruker ARX 300 instrument at room temperature with deuterated chloroform as the solvent for *azido-poly(PEGMA)* and deuterated water for *alkynyl-poly(META)*. GPC was performed on a Waters GPC system, equipped with a Waters 1515 isocratic HPLC pump, a Waters 717 plus Autosampler injector, and a Waters 2414 refractive index detector. The eluent was dimethylformamide (DMF) for *azido-poly(PEGMA)* at a flow rate of $1.0 \text{ mL}\cdot\text{min}^{-1}$, and the calibration curve was generated using polystyrene molecular weight standards. For *alkynyl-poly(META)*, the eluents was aqueous sodium nitrate (NaNO_3) solution ($0.1 \text{ mol}\cdot\text{l}^{-1}$) at a flow rate of $1.0 \text{ mL}\cdot\text{min}^{-1}$ and the calibration curve was generated using PEG standards.

2.5. Surface Characterization. Chemical composition of the copolymers and polymer-functionalized substrates was determined by X-ray photoelectron spectroscopy (XPS). XPS measurements were carried out on a Kratos AXIS Ultra spectrometer with a monochromatized Al K α X-ray source (1486.6 eV photons), at a constant dwelling time of 100 ms and pass energy of 40 eV. All binding energies (BEs) were referenced to the neutral C 1s hydrocarbon peak at 284.6 eV. The topography of the polymer-modified substrate surfaces was investigated in a dry state by atomic force microscope (AFM), using a Nanoscope IIIa AFM from Digital

Instruments, Inc. The root-mean-square (rms) roughness (R_s) of the surfaces was calculated from the roughness profile determined by AFM. To measure the thickness of multilayer polymer coatings, the polymer-functionalized substrates were scratched using a razor blade.^{40,41} As the glass substrate is much harder than the polymer layer, only the polymer coatings are scratched away, and the thickness of polymer layer can be imaged. The thicknesses of multilayer polymer coatings were also determined by ellipsometry. The measurements were carried out on a variable angle spectroscopic ellipsometer (model VASE, J.A. Woollam Inc., Lincoln, NE) at incident angles of 65° and 75° in the wavelength range 500–1000 nm. Data were recorded and processed using the WVASE32 software package.

2.6. Antifouling Assays. Gram-negative *Pseudomonas sp.* NCIMB 2021 bacteria were used to evaluate the antibacterial adhesion characteristics and bactericidal efficacy of the polymer multilayer coatings. *Pseudomonas sp.* NCIMB 2021 was cultured in a nutrient-rich artificial seawater medium as described previously.⁴² After incubation, the bacterial suspension was centrifuged at 2700 rpm, and the supernatant was removed. Bacterial cells were washed with artificial seawater twice and resuspended at a concentration of 10^7 cells/mL. Each substrate was then immersed in 1 mL of bacterial suspension under static condition at 30 °C for 24 h. The live/dead two-color fluorescence method was used to assess the bactericidal effects of polymer-functionalized SS substrates.³⁹ By staining the bacterial cells with LIVE/DEAD BacLight Bacterial Viability Kit, the viable (appearing green) and dead (appearing red) bacterial cells can be distinguished under a fluorescent microscopy. Adhered bacterial cells were investigated under a scanning electron microscope (SEM) (JSM-5600 model, JEOL Co., Tokyo, Japan) after fixing with 3% glutaraldehyde and dehydration with serial ethanol. Quantification of bacteria adhesion and viability on the pristine and polymer-modified SS was carried out by the spread plate method.³⁹

Barnacle cyprids were cultured by published methods.^{43,44} One milliliter of filtered seawater containing about 50 barnacle cyprids were added to the pristine glass vials and glass vials modified by polymer multilayer coatings. Three replicates were used for each assay. After 24 h of incubation, the dead barnacles and barnacle cyprids settled in each vial were counted under a stereoscope.

3. RESULTS AND DISCUSSIONS

3.1. Synthesis and Characterization of Clickable Polymers. To take advantage of click chemistry for covalent LBL assembly, clickable azide and alkyne groups were incorporated in the side-chains of functional polymer backbones. Low-fouling clickable *azido-poly(PEGMA)* chains were synthesized in two steps. The random copolymer, (poly-(PEGMA-co-GMA)), was first prepared by RAFT polymerization. Subsequent reaction of sodium azide with the pendant oxirane rings in the GMA units of the copolymer introduced the azide groups.³⁵

The respective FTIR spectra of the poly(PEGMA-co-GMA) and *azido-poly(PEGMA)* copolymers are shown in Figure 1a,b. The absorption bands at the wavenumbers of about 1728 cm^{-1} and 1095 cm^{-1} , associated with ester and ether stretching, respectively,⁴⁵ are present in both poly(PEGMA-co-GMA) and *azido-poly(PEGMA)*. After reaction with sodium azide, a new absorption band appears at the wavenumbers of 2106 cm^{-1} , corresponding to the vibration frequency of the pendant azide groups in *azido-poly(PEGMA)*.^{35,45} The successful introduction of azide groups in *azido-poly(PEGMA)* was also confirmed by XPS analysis (Figure 2a–c) and ^1H NMR spectroscopy (Figure 3a). Figure 2a–c show the XPS wide scan, C 1s, and N 1s core-level spectra of *azido-poly(PEGMA)*. The XPS N 1s core-level spectrum (Figure 2c) is curve-fitted into three peak components with BEs at 399.9, 400.6, and 403.9 eV, respectively, associated with the negatively charged nitrogen

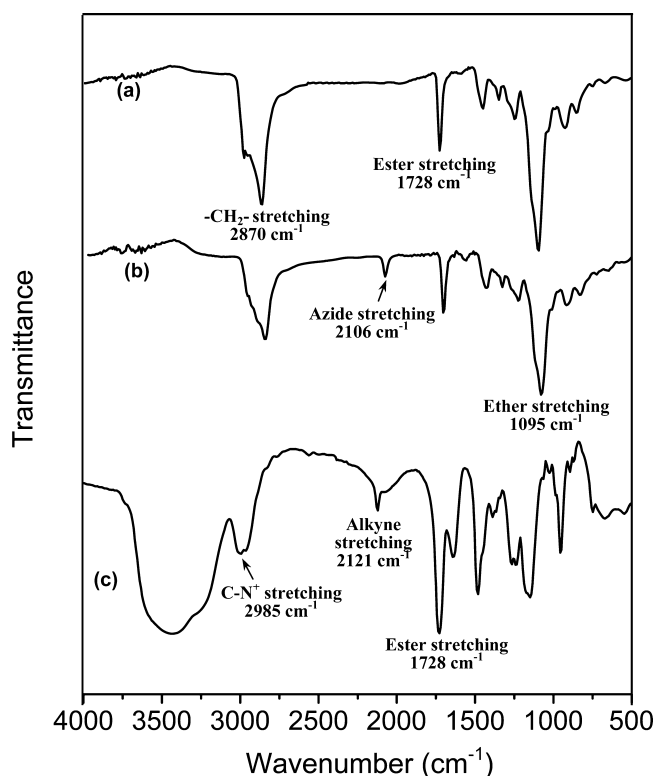


Figure 1. FTIR spectra of (a) poly(PEGMA-co-GMA), (b) azido-poly(PEGMA), and (c) alkynyl-poly(META).

(N⁻), the imine nitrogen ((N)-N=), and the positively charged nitrogen (N⁺) of the azide groups.^{35,46} The molar ratio of the three species is about 1:1:1, as determined from the spectral area ratio of the three peak components, in agreement with the theoretical ratio of azide structure ($-\text{N}=\text{N}^+=\text{N}^-$). The molar fraction of azide components in azido-poly-

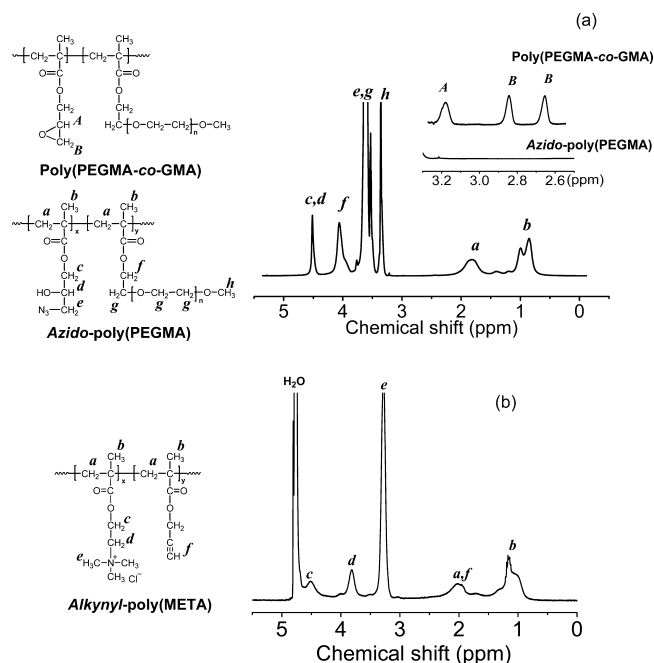


Figure 3. ¹H NMR spectra of (a) poly(PEGMA-co-GMA) and azido-poly(PEGMA), and (b) alkynyl-poly(META).

(PEGMA) ($[\text{azide units}]/([\text{azide units}] + [\text{PEGMA units}])$) is about 14.5 mol %, as calculated from [N]/[C] and [N]/[O] molar ratios derived from the XPS C, O, and N 1s core-level spectral area ratios.

Figure 3a shows the ¹H NMR spectra of poly(PEGMA-co-GMA) and azido-poly(PEGMA). The chemical shifts at 3.20 ppm (A, CH-O of the oxirane ring), and 2.82 and 2.62 ppm (B, CH₂-O of the oxirane ring) in poly(PEGMA-co-GMA) has disappeared completely in azido-poly(PEGMA),^{47,48} indicating the highly efficient oxirane ring-opening reaction by sodium

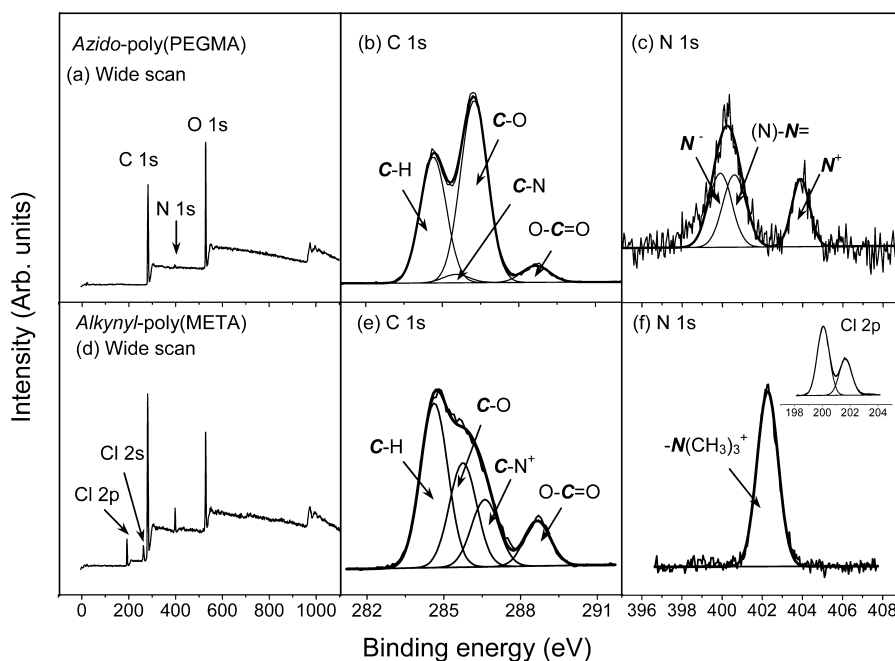


Figure 2. XPS wide scan, C 1s, and N 1s core-level spectra of (a–c) azido-poly(PEGMA) and (d–f) alkynyl-poly(META). Inset of f: XPS Cl 2p core-level spectrum of alkynyl-poly(META).

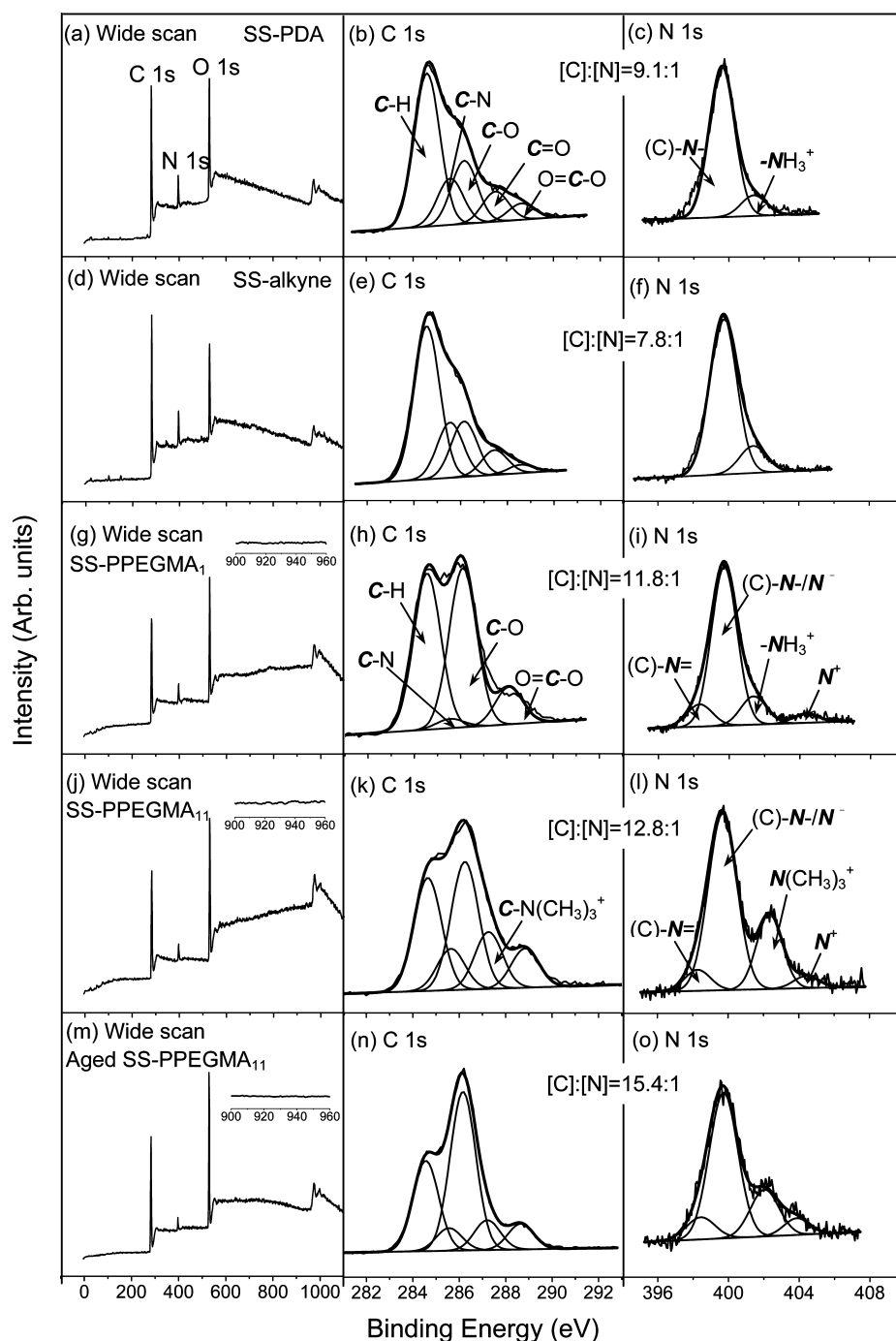


Figure 4. XPS wide scan, C 1s, and N 1s core-level spectra of the (a–c) polydopamine-coated SS (SS-PDA), (d–f) SS-alkyne, (g–i) SS-PPEGMA₁, (j–l) SS-PPEGMA₁₁ and (m–o) aged SS-PPEGMA₁₁ surfaces. (Aged substrate: immersion in filtered seawater for 30 days).

azide. Concomitantly, a new chemical shift at $\delta = 4.5$ ppm (c,d), associated with the COOCH_2 and CH-OH species in *azido*-poly(PEGMA), has appeared.^{35,47,48} The chemical shifts at $\delta = 3.7$ ppm (g), 4.1 ppm (f), and 3.3 ppm (h) are attributable to the OCH_2 , COOCH_2 , and OCH_3 groups, respectively, in the side chain of PEGMA units in *azido*-poly(PEGMA).^{47,49} The molar ratio of PEGMA units to the GMA units in poly(PEGMA-co-GMA) was about 5.4:1, as calculated from the ratio of integrals of the resonance at 3.3 ppm (OCH_3 in PEGMA unit) to that at 2.62 ppm (B, $\text{CH}_2\text{-O}$ of the oxirane ring). Accordingly, *azido*-poly(PEGMA) should contain about 15.6 mol % azide functional groups, based on the

complete oxirane ring-opening reaction. This result reveals that the PEGMA macromonomer polymerizes faster than GMA in the copolymerization process. A similar phenomenon was observed in the RAFT copolymerization of oligo(ethylene glycol) methyl ether methacrylate (OEGMA₁₁₀₀) macromonomer and di(ethylene glycol) methyl ether methacrylate (DEGMA).⁵⁰ The high reactivity of macromonomers might be due to nonideal mixing effects, which can enhance the propagation rate and limit termination between compartmentalized propagating radicals.^{50,51} The XPS and ^1H NMR results are in agreement and indicate that all of the oxirane rings in the

poly(PEGMA-co-GMA) copolymer have been converted to azide groups.

The *alkynyl*-poly(META) copolymer was prepared by conventional free radical copolymerization of META and PMA, and was also characterized by FTIR spectroscopy, XPS, and ^1H NMR spectroscopy. As shown in Figure 1c, the characteristic absorption bands at the wavenumbers of about 2985 cm^{-1} , 2121 cm^{-1} , and 1728 cm^{-1} are associated with the stretching vibration of $\text{C}-\text{N}^+$, alkyne, and ester groups, respectively,^{45,52} indicating the incorporation of QACs and alkyne groups in the copolymer. The XPS N 1s core-level spectrum of *alkynyl*-poly(META) (Figure 2f) is dominated by positively charged nitrogen species at the BE of 402.1 eV, consistent with the presence of QACs ($-\text{N}(\text{CH}_3)_3^+$) in *alkynyl*-poly(META).^{46,53} The molar fraction of alkyne components in *alkynyl*-poly(META) ($[\text{alkyne units}]/([\text{alkyne units}] + [\text{META units}])$) is around 40.2%, as calculated from $[\text{N}]/[\text{C}]$ and $[\text{N}]/[\text{O}]$ molar ratios derived from the XPS C, O, and N 1s core-level spectral area ratios (Figure 2d–f). The chemical shift at $\delta = 3.3\text{ ppm}$ (e) in the ^1H NMR spectrum of Figure 3b is attributable to the CH_3 in the quaternary ammonium groups.^{47,54} The broad peak in the range of 1.8–2.2 ppm (a,f) is associated with the CH_2C groups of the polymer backbone and $\text{C}\equiv\text{CH}$ in the alkyne groups.^{47,55} Therefore, the spectroscopic results indicate that clickable *alkynyl*-poly(META) has been successfully synthesized for the subsequent LBL click assembly.

3.2. Antifouling and Antibacterial Polymer Multilayer Coatings on SS. To covalently couple the clickable copolymer chains on the substrates, the alkyne groups were first introduced on the SS surface as the prelayer via the Schiff-base and Michael addition reaction of propargylamine with the SS-PDA.³⁸ The successful introduction of alkyne functionality on the substrates (SS-alkyne) was confirmed by XPS analysis (Figure 4a–f). For the SS-PDA and SS-alkyne surfaces, the XPS C 1s core-level spectra (Figure 4b,e) are curve-fitted into five peak components with BEs at 284.6, 285.6, 286.2, 287.4, and 288.5 eV, attributable to the $\text{C}-\text{H}$, $\text{C}-\text{N}$, $\text{C}-\text{O}$, $\text{C}=\text{O}$ and $\text{O}-\text{C}=\text{O}$ species, respectively;^{39,46} the XPS N 1s core-level spectra (Figure 4c,f) are curve-fitted into two peak components with BEs at 399.7 and 401.5 eV, associated with the amine ($((\text{C})-\text{N}-)$) and protonated amine (NH_3^+) species, respectively. The intensity of $\text{C}-\text{N}$ peak component has increased after coupling of propargylamine on the SS-PDA surface. Also, the $[\text{C}]/[\text{N}]$ molar ratio, calculated from the XPS C 1s and N 1s core-level spectral area ratio, has decreased from 9.1:1 to 7.8:1, consistent with a lower $[\text{C}]/[\text{N}]$ ratio of propargylamine (3:1) than that of the dopamine (8:1). Apart from the reduction in $[\text{C}]/[\text{N}]$ ratio, the decrease in the molar fraction of oxygen ($([\text{O}]/([\text{C}] + [\text{O}] + [\text{N}]))$) on the surface (from 21.7 mol % to 16.7 mol %) also confirms the coupling of propargylamine.

The LBL click deposition process starts with the alkyne–azide cycloaddition of *azido*-poly(PEGMA) to the SS-alkyne surface to form the first low-fouling layer. Figure 4g–i shows the XPS wide scan, C 1s, and N 1s spectra of the *azido*-poly(PEGMA) clicked SS-alkyne surface (denoted as SS-PPEGMA₁). In comparison to the wide scan spectrum of SS-alkyne surface (Figure 4d), the oxygen signal at the BE of about 532 eV has increased significantly ($([\text{O}]/([\text{C}] + [\text{O}] + [\text{N}]))$: from 16.7 mol % to 25.1 mol %) in the wide scan spectrum of SS-PPEGMA₁ surface (Figure 4g), consistent with the high oxygen content of *azido*-poly(PEGMA). The XPS C 1s core-level spectrum (Figure 4h) is curve-fitted into four peak

components with BEs at 284.6, 285.6, 286.2, and 288.1 eV, attributable to the $\text{C}-\text{H}$, $\text{C}-\text{N}$, $\text{C}-\text{O}$, and $\text{O}-\text{C}=\text{O}$ species, respectively.^{26,46} The intensity of the $\text{C}-\text{O}$ peak component in the XPS C 1s core-level spectrum of the SS-PPEGMA₁ surface has increased substantially (Figure 4h). These results indicate the successful coupling of *azido*-poly(PEGMA) to the SS-alkyne surface via click reaction. It is expected that only part of the clickable azide functional groups in *azido*-poly(PEGMA) have been consumed, leaving behind some free azide groups for the subsequent LBL click assembly. The presence of unreacted free azide groups on the surface was confirmed by the XPS N 1s core-level spectrum of the SS-PPEGMA₁ surface (Figure 4i). The N 1s core-level spectrum can be curve-fitted into four peak components.^{35,46} The peak component at 399.7 eV is attributed to the amine ($((\text{C})-\text{N}-)$) moiety in the SS-alkyne layer and the triazole rings, as well as the residual negatively charged azide nitrogen (N^-). The peak component with BE at 401.4 eV is assigned to the protonated amine (NH_3^+) species from the underlying SS-alkyne layer. The other two peak components, with BEs at 398.3 and 404.3 eV, are attributable to the imine ($((\text{C})-\text{N}=\text{N})$) species of the triazole rings and positively charged nitrogen (N^+) of the unreacted azide groups, respectively. The molar ratio of the imine ($((\text{C})-\text{N}=\text{N})$) moiety in the triazole rings and the positively charged nitrogen (N^+) in the residual azide group, is around 2.6:1, indicating that about 56 mol % of the azide groups were consumed in the click reaction, and 44 mol % were available for the subsequent LBL click assembly.

After clicking 11 alternative layers of polymer chains on the substrate, the XPS C 1s core-level spectrum of the SS-PPEGMA₁₁ surface (Figure 4k) can be curve-fitted into five peak components with BEs at 284.6, 285.5, 286.3, 287.2, and 288.7 eV, attributable to the $\text{C}-\text{H}$, $\text{C}-\text{N}$, $\text{C}-\text{O}$, $\text{C}-\text{N}(\text{CH}_3)_3^+$ and $\text{O}-\text{C}=\text{O}$ species.^{26,46} The significant increase in intensity of the $\text{C}-\text{O}$ species and the appearance of $\text{C}-\text{N}(\text{CH}_3)_3^+$ species indicate the successful immobilization of *azido*-poly(PEGMA) and *alkynyl*-poly(META) on the surface via click reaction. The XPS N 1s core-level spectrum (Figure 4l) comprises four peak components. Three peak components, with respective BEs at 398.3, 399.7, and 404.4 eV, are associated with the imine ($((\text{C})-\text{N}=\text{N})$) species of the triazole rings, amine nitrogen ($((\text{C})-\text{N}-)$) and positively charged nitrogen (N^+) in residual azide groups.^{35,46} The fourth peak component with BE at 402.3 eV is attributable to the positively charged nitrogen of QACs ($\text{N}(\text{CH}_3)_3^+$), comprising about 25 mol % of all the nitrogen species. The polymer multilayer coatings are not necessarily highly ordered as shown in Scheme 1.²¹ The two layers of polymer chains are intertwined, leading to the presence of a small quantity of QACs on the SS-PPEGMA_x surfaces and low-fouling PEGMA units on the SS-PMETA_x surfaces. In addition, the free azide groups on the SS-PPEGMA₁₁ surface can be employed for further functionalization.

The stability of the LBL-functionalized surface was investigated by exposure to filtered (0.2 μm) natural seawater at 30 °C for 30 days. The XPS analysis of the aged SS-PPEGMA₁₁ surface composition is shown as Figure 4m–o. The XPS C 1s and N 1s core-level spectral line shapes do not change appreciably after exposure to the seawater for 30 days, indicating the stability and reliability of the multilayer polymer coatings prepared by LBL click deposition. It is important to ensure the complete removal of residual copper catalyst on the surfaces after the LBL deposition process, as copper could play a role in the observed antifouling effects. No copper signals (Cu

$2p_{3/2}$ and Cu $2p_{1/2}$ in the BE range of 900–960 eV) are discernible in the XPS wide scan spectra of SS-PPEGMA₁₁, SS-PPEGMA₁₀, and aged SS-PPEGMA₁₁ surfaces (Figure 4g,j,m and insets). The results indicate that no copper residues were presented on the LBL deposited surfaces to interfere with the subsequent antimicrobial and antifouling assays.

The whole LBL click assembly process was investigated by XPS. Figure 5 shows the XPS C 1s and N 1s core-level spectra

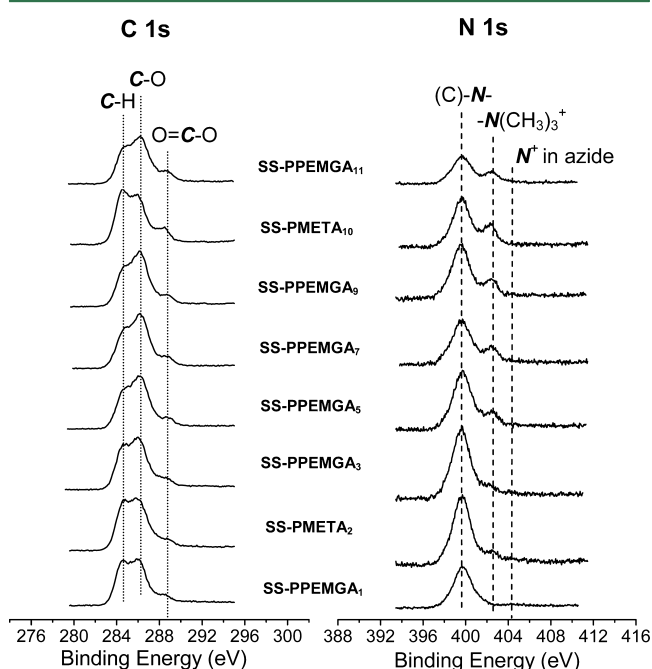


Figure 5. XPS C 1s and N 1s core-level spectra of the SS substrates functionalized with increasing number of alternating antifouling azido-poly(PEGMA) and antimicrobial alkynyl-poly(META) layers by LBL click deposition.

of surfaces functionalized with a progressive number of polymer layers. Briefly, the C 1s core-level spectra consist of three peak components with BEs at 284.6, 285.6, and 288.4 eV, attributable to C–H, C–O, and O–C=O species, respectively. After clicking of each layer of azido-poly(PEGMA) on the surface, the intensity of C–O peak component at the BE of 286.2 eV increases, consistent with the presence of a poly(PEGMA) outer layer. The N 1s core-level spectra consist predominantly of three peak components with BEs at about 399.7, 402.3, and 404.3 eV, attributable to the amine ((C)–N–) moiety, positively charged ammonium nitrogen ($-N(CH_3)_3^+$), and positively charged nitrogen (N^+) of the azide groups. The intensity of positively charged ammonium nitrogen ($-N(CH_3)_3^+$) at 402.3 eV, corresponding to the QACs in alkynyl-poly(META),^{26,46} increases gradually with the number of layers. Therefore, the successful preparation of polymer multilayer brush coatings via LBL click assembly has been ascertained.

The topology of the multilayer polymer-functionalized surfaces was investigated by AFM, as shown in Figure 6. The rms roughness of the polymer-functionalized surfaces decreases from 2.06 to 1.10 nm over an area of $1 \times 1 \mu\text{m}$, with increasing number of polymer layers on the surface. The result implies the formation of a more complex and dense polymer coating on the surface (Figure 6). The thicknesses of the polymer multilayer coatings were determined by ellipsometry and AFM analysis of

the scratched surfaces. For AFM scratch analysis, the change in phase signals between the scratched area and nonscratched area is indicative of regions with different composition⁵⁵ (Figure 6 and Figure S1, Supporting Information). The AFM phase images thus reveal that the scratches have reached the substrates. The thickness of coating increases by about 60 nm after 11 polymer layers have been clicked onto the surface (Figure 7), implying an average bilayer thickness of about 12 nm. Taking the bulky PEGMA side chains into account, it is reasonable that the azido-poly(PEGMA) and alkynyl-poly(META) bilayer is thicker than the poly(acrylic acid) bilayer (4.2–4.6 nm) reported before.^{20,22} In addition, the thickness for the polymer bilayer reduces gradually with the number of deposited layers, as evidenced by the larger average thickness for the first two bilayers (~ 14 nm) than that of last three bilayers (~ 10 nm) (Figure 7). The reduction in bilayer thickness probably arises from the less clickable azide or alkyne functional groups on the surfaces in the later LBL click deposition process.⁵⁶

3.3. Bacterial Fouling Assays of the LBL Click-Deposited Surfaces. The antibacterial fouling efficacy of the polymer multilayer coatings was evaluated against the Gram-negative marine bacteria, *Pseudomonas sp.* NCIMB 2021. Figure 8 shows the SEM images of surfaces after bacterial fouling. The pristine SS surface is highly susceptible to bacterial adhesion and colonization. A large amount of bacterial cells adhered readily on the surface, either individually or in small clusters (Figure 8a). Most bacterial cells are rod-shaped and polar flagellated, with a length of 1.0–2.5 μm .⁵⁷ For the LBL-functionalized SS surfaces, the antibacterial adhesion performance is correlated with the number of polymer layers clicked on the substrates. For the SS-PPEGMA₁ and SS-PMETA₂ surfaces (Figure 8b,c), an obvious reduction in bacterial adhesion was observed as compared to that of the pristine SS. The SS-PPEGMA₅ surface exhibited a higher efficiency in resisting bacterial attachment (Figure 8d) than the SS-PPEGMA₁ and SS-PMETA₂ surfaces. Only a few bacterial cells were present on the SS-PMETA₁₀, and almost no bacterial cells could be found on the SS-PPEGMA₁₁ surface (Figure 8e,f), indicating superior functionality of the denser polymer multilayer coatings. The resistance to bacterial adhesion of the spin-coated SS-Mix surface was not as good as the of SS-PPEGMA₁₁ surface, as evidenced by the presence of a small number of bacterial cells on the surface (Figure 8g).

The bactericidal efficacy of functionalized SS surfaces was also assayed using the live/dead two-color fluorescence method. Figure 9 shows fluorescence microscopy images of pristine and LBL-functionalized SS surfaces after exposure to *Pseudomonas sp.* NCIMB 2021 for 24 h. After staining with the combination dye, the viable cells appear green, while the dead cells appear red under the fluorescence microscope. A high concentration of viable *Pseudomonas sp.* NCIMB 2021 cells (stained green) adhered on the pristine SS surface (Figure 9a), with very few dead cells (stained red) (Figure 9b), indicating the high susceptibility of pristine SS to bacterial adhesion and colonization. In comparison to the pristine SS surface, the amount of viable bacterial cells attached to the SS-PMETA₁₀ and SS-PPEGMA₁₁ surfaces has decreased substantially (Figure 9c,e), indicating high antibacterial efficiencies of the multilayer polymer coatings. Some dead bacterial cells were observed on the SS-PMETA₁₀ surface (Figure 9d), suggesting bactericidal efficacy of the poly(META) layer. In addition, the presence of some dead cells on the SS-PPEGMA₁₁ surface (Figure 9f)

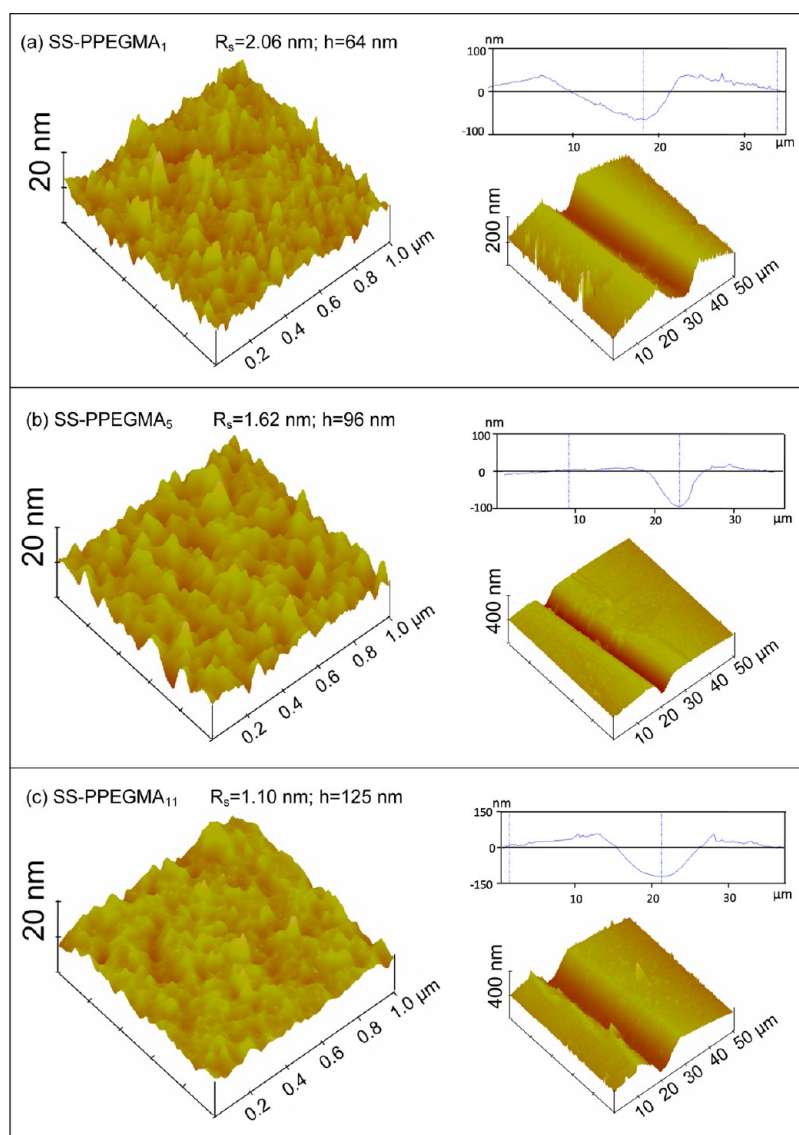


Figure 6. AFM images of the (a) SS-PPEGMA₁, (b) SS-PPEGMA₅, and (c) SS-PPEGMA₁₁ surfaces. R_s = rms roughness, h = thickness of polymer multilayer coating.

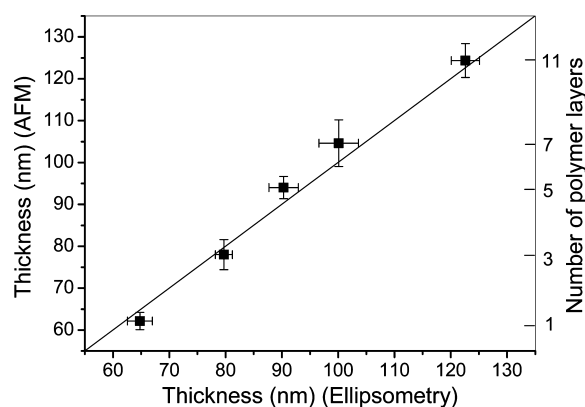


Figure 7. Thickness of polymer multilayer coatings measured by ellipsometry and AFM scratch analysis.

indicates that the alternating polymer layers become intertwined with an increasing number of layers. For the spin-coated SS-Mix surface, more dead cells and less viable cells

were observed in comparison to that on the pristine SS surface. The antibacterial performance improves with increasing number of polymer layers, as evidenced by the presence of less viable cells on the SS-PMETA₁₀ and SS-PPEGMA₁₁ surfaces (Figure 9c–f) than that on the SS-PPEGMA₁, SS-PMETA₂, and SS-PPEGMA₅ surfaces (Figure S2a–f, Supporting Information).

Quantitative determination of the number of viable bacterial cells on the multilayer polymer-functionalized surfaces was conducted by the spread plate method (Figure 10). Bacterial adhesion on all the polymer-functionalized surfaces was greatly reduced by 97% from that of the pristine SS surface, with the SS-PPEGMA₁₁ surface exhibiting the highest antibacterial efficacy. These results indicate the superior antibacterial performance of the multilayer polymer coatings. These results are also consistent with those obtained from SEM and fluorescence microscopy images. The formation of thicker and denser polymer coating helps to account for the higher antibacterial efficacy of the surfaces. The enhanced antiadhesion and bactericidal efficiency of the multilayer polymer coatings probably arises from the large increase in excluded volume of

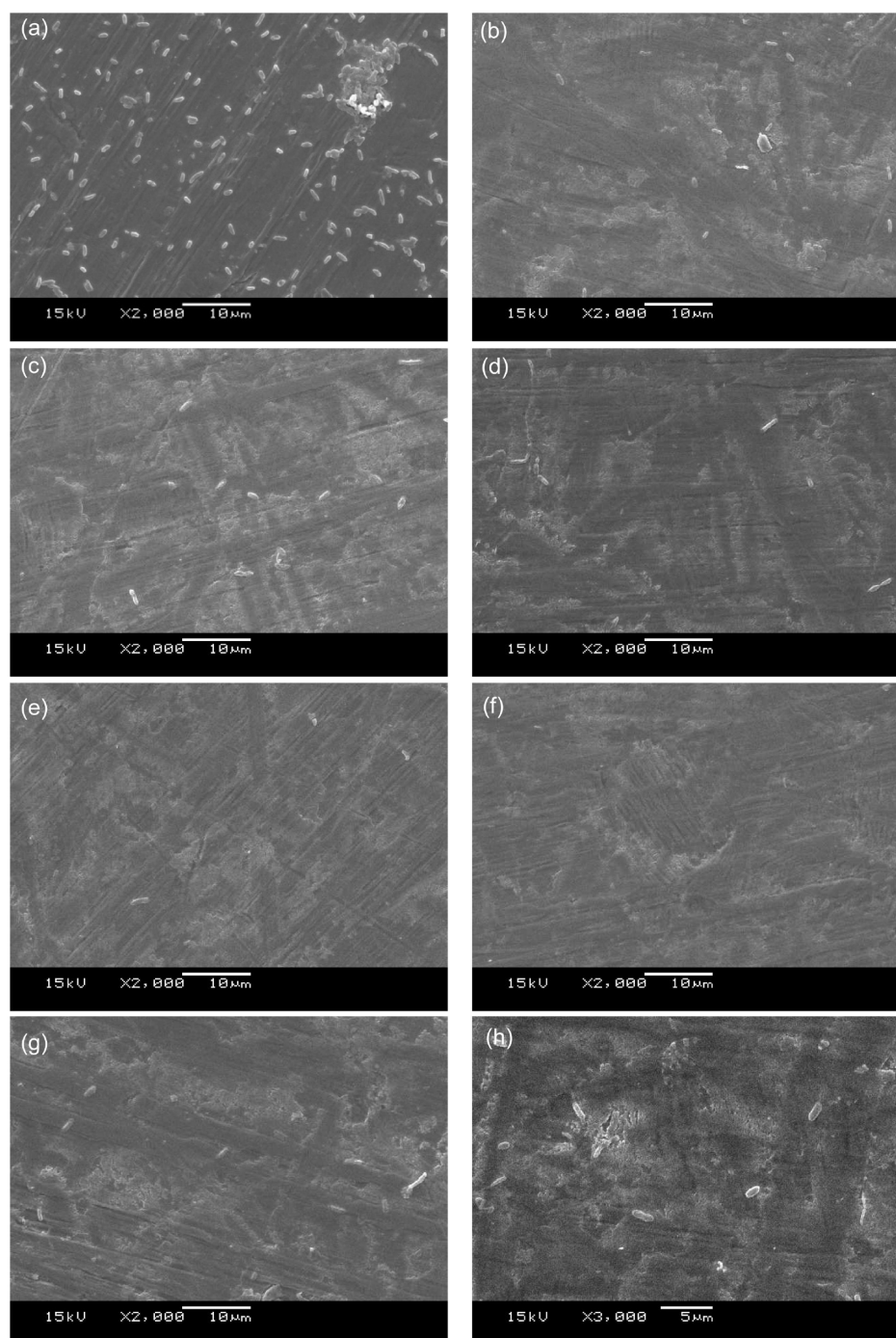


Figure 8. SEM images of the (a) pristine SS, (b) SS-PPEGMA₁, (c) SS-PMETA₂, (d) SS-PPEGMA₅, (e) SS-PMETA₁₀, (f) SS-PPEGMA₁₁, (g) SS-Mix, and (h) aged SS-PPEGMA₁₁ surfaces after exposure to *Pseudomonas sp.* NCIMB 2021 (10^7 cells/ml) for 24 h. (Aged substrate: immersion in filtered seawater for 30 days).

the highly hydrated poly(PEGMA) chains²⁹ in the multilayer and the disruption of microbial membrane by the poly(META) chains.⁵⁸

The antibacterial efficiency of the multilayer polymer coatings was investigated after the substrates were exposed to filtered ($0.2\ \mu\text{m}$) natural seawater at $30\ ^\circ\text{C}$ for 30 days. Only a small number of adherent bacteria were observed on the 30-day aged SS-PPEGMA₁₁ surface, as evidenced by the SEM images in Figure 8h. The number of viable bacteria on the aged SS-PPEGMA₁₁ surface increased only slightly (Figure 9i), in comparison to that of the as-prepared SS-PPEGMA₁₁ surface.

The presence of dead bacterial cells indicates the persistence of antimicrobial efficacy of the aged SS-PPEGMA₁₁ surface (Figure 9j). Thus, the stability and durability of the multilayer polymer coatings prepared by LBL click deposition have been ascertained. Apart from the chemical stability, the mechanical property of the multilayer polymer coatings has been studied in a simple abrasion test. Figure S3 (Supporting Information) shows the respective AFM topographical image of the abraded SS-PPEGMA₁₁ surface and SEM image of the surface after bacterial fouling. The rms roughness decreased from 1.1 to 0.3 nm. The antibacterial efficacy was compromised, indicating the

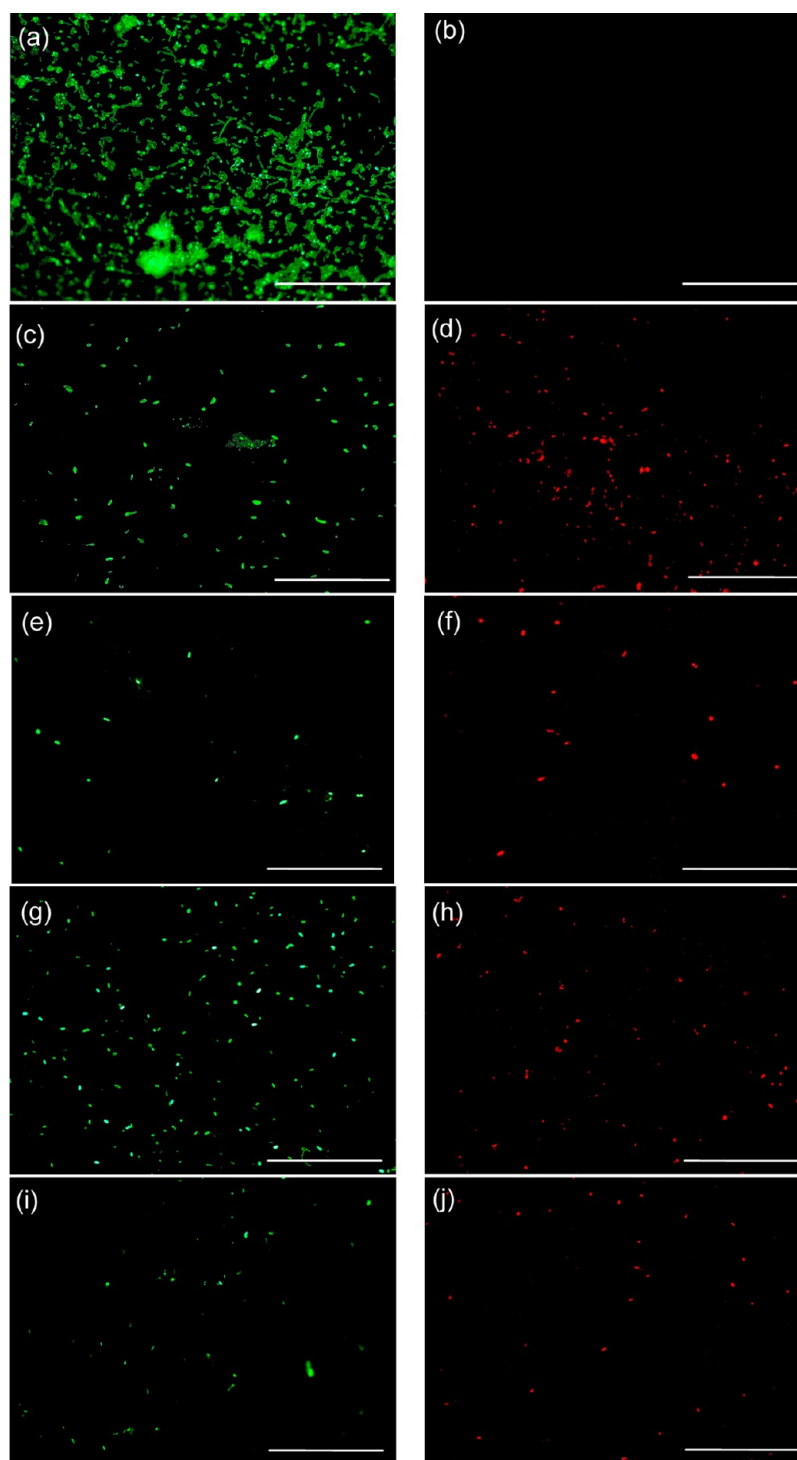


Figure 9. Fluorescence microscopy images of the (a,b) pristine SS, (c,d) SS-PMETA₁₀, (e,f) SS-PPEGMA₁₁, (g,h) SS-Mix, and (i,j) aged SS-PPEGMA₁₁ surfaces after exposure to *Pseudomonas* sp. NCIMB 2021 (10^7 cells/ml) for 24 h. Scale bar: 50 μ m. (Aged substrate: immersion in filtered seawater for 30 days).

poor mechanical property of the LBL click-deposited polymer coatings.

3.4. Barnacle Cyprid Settlement Assays of the LBL Click-Deposited Surfaces. Barnacles are one of the most problematic macrofoulers with a wide geographical distribution.^{43,44,59} In this work, settlements of barnacle cyprids were used to evaluate the antifouling efficacies of the polymer multilayer coatings from LBL click deposition. Figure 11 shows the results of settlement assays on the pristine and polymer-

functionalized glass surfaces after an exposure time of 24 h. About 42% of the barnacle cyprids settled on the pristine glass substrate (GS) surfaces. Settlement was reduced to 21% on the GS-PPEGMA₁ surface. Click deposition of *alkynyl*-poly-(META) polymer as the second layer decreased the settlement fraction to 19%. After coupling of two additional bilayers, only 9.2% of the barnacle cyprids settled on the GS-PPEGMA₅ surface, indicating an increase in efficiency with the number of polymer layers clicked on the surface. However, the settlement

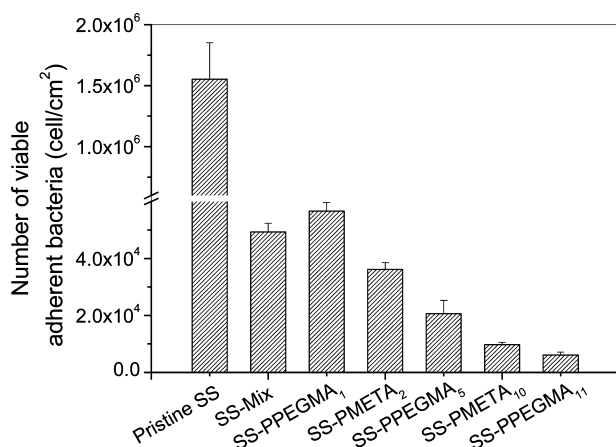


Figure 10. Number of viable adherent bacterial cells after exposure to *Pseudomonas* sp. NCIMB 2021 (10^7 cells/ml) for 24 h. The cell number was determined by the spread plate method. Each error bar represents the standard deviation calculated from three replicates.

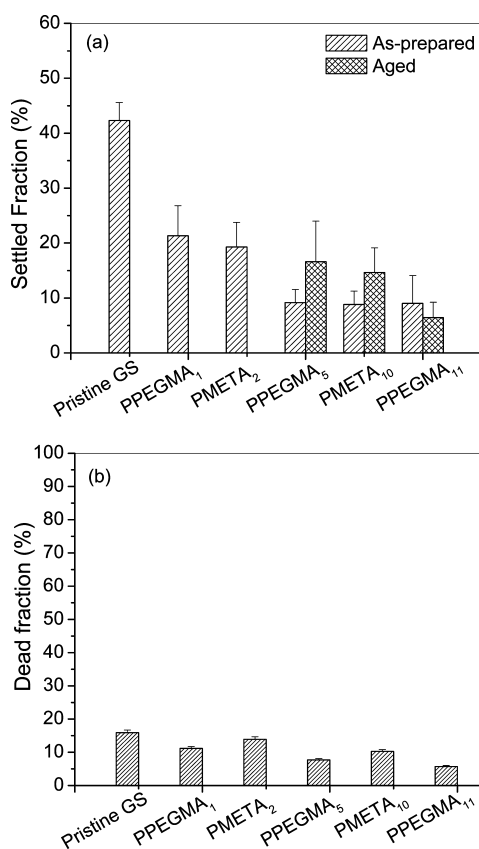


Figure 11. (a) Settled fractions of barnacle cyprids on the pristine and LBL click-assembled substrates after 24 h of incubation (aged substrate: immersion in filtered seawater for one week). (b) Dead fractions of barnacle cyprids on the pristine and LBL click-assembled substrates after 24 h of incubation.

remained at 9% with further increase in the number of layers to 11 for GS-PMETA₁₀ and GS-PPEGMA₁₁ surfaces.

To further compare the antifouling performance of the GS-PPEGMA₅, GS-PMETA₁₀, and GS-PPEGMA₁₁ surfaces, the three surfaces were aged in filtered (0.2 μ m) natural seawater for 1 week and then tested again for barnacle settlement. The three surfaces still exhibit resistance to settlement. About 16% and 14% of the barnacles settled on the aged GS-PPEGMA₅

and GS-PMETA₁₀ surfaces. The fraction of settlement on the aged GS-PPEGMA₁₁ surface was 6.4%, implying stability and thus higher antifouling efficiency of the surface with more polymer layers over an extended period of time. The dead fractions for polymer-functionalized surfaces were between 5% and 15% after the cyprids were in contact with the respective surfaces for 24 h, which is lower or comparable to that for the pristine GS surface (~15%). These results indicate that no toxic copper residues were presented on the surface. The resistance to barnacle cyprid settlement can be attributed to the low-fouling and antibacterial polymer coatings. These robust polymer multilayer coatings of low toxicity suggest their useful applications as environmentally friendly antifouling coatings.

4. CONCLUSIONS

By combining the LBL deposition technique and click chemistry, a unique approach to fabricating bifunctional and robust polymer multilayer coatings was developed to combat marine biofouling. The dense and stable bifunctional polymer multilayer coatings were prepared by LBL click assembly of the low-fouling azido-poly(PEGMA) and bactericidal alkynyl-poly(META) via Huisgen 1,3-dipolar cycloaddition. The so-prepared polymer multilayer coatings exhibit good resistance to bacterial adhesion and high bactericidal efficiency against marine *Pseudomonas* sp. NCIMB 2021. The settlement of barnacle (*Amphibalanus* (=Balanus) amphitrite) cyprids was reduced by up to 80% on the polymer multilayer coatings. The antifouling efficiency against both bacterial adhesion and barnacle cyprids settlement improves with the number of multilayers. The polymer multilayer coatings also exhibit low toxicity to barnacle cyprids and are stable upon prolonged exposure to seawater. The polymer multilayer coatings are thus potentially useful as effective and environmentally benign antifouling coatings.

■ ASSOCIATED CONTENT

Supporting Information

AFM topographies of the scratched SS-PPEGMA₁, SS-PPEGMA₅, and SS-PPEGMA₁₁ surfaces, fluorescence microscopy images of SS-PPEGMA₁, SS-PMETA₂, and SS-PPEGMA₅ surfaces after bacterial fouling, and experimental details and results on the abrasion test of the multilayer polymer coating. This material is available free of charge via the Internet at <http://pubs.acs.org>.

■ AUTHOR INFORMATION

Corresponding Author

*E-mail address: cheket@nus.edu.sg (E.T.K.); tmsteolm@nus.edu.sg (S.L.-M.T.); ritt@duke.edu (D.R.).

Notes

The authors declare no competing financial interest.

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